

AI-Based Student Performance & Dropout Prediction System: Machine Learning, Individual Risk Scoring and Analytics for Early Academic Intervention

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Abstract—Monitoring of students in many educational institutions happens with the help of attendance registers and internal marks sheets that are considered only after the appearance of the problem. This paper introduces the Student Performance and Dropout Prediction System called EDU AI, which uses the data about attendance, grades for assignments and quizzes, marks in exams, study time, GPA, demographics, extracurricular activities and dropout cases. There are four machine learning algorithms with specific tasks: Logistic Regression for performance estimation, Random Forest for identifying the risk of being dropped out, Naïve Bayes for evaluating the risk of the behavior and K-Means for student segmentation. The output of these models is combined into the risk score for each student (Low - less than 50%, Medium - from 50 to 80%, High - more than 80%) with the description of influencing factors and visualized via faculty dashboard. For the dataset of 248 students, the best results were obtained by the model of Random Forest (accuracy 94.8%, precision 92.1%, recall 94.5%, F1-score 93.3%). In addition, the second place took the Logistic Regression with 91.2% accuracy. The system found 41 risky students, among which 12 belong to the group of critical ones. Among the confusion matrices of Random Forest, there was a case of one High-risk student who was classified as Low-risk. These results are achieved using a small dataset of unknown origin and validation.

Keywords— student performance prediction, dropout prediction, educational data mining, machine learning, random forest, risk scoring, early warning, learning analytics

I. INTRODUCTION

The result is dependent on various interrelated variables such as classroom attendance, consistent submission of tasks, success in quizzes and exams, time spent studying on their own, past academic performance, and individual and co-curricular activities background. When various variables start faltering, it could mean that the student is already moving towards failure or toward dropping out of the program. Dropping out is detrimental not only to the student but also to the institution because their success will be based on it.

In most organizations, progress can be tracked via

attendance lists, internal assessments and semester-end results. The data can be useful but is generally used when there is failure in examination or long-term absence. The warnings may be individual, such as reduced attendance, reduced test score or reduced hours spent studying, and the overall impact on an individual student may not be detected by the teacher because of the number of students that have to be tracked simultaneously. The current reports are more of a descriptive nature and do not make any predictions about what will happen or prioritize the students according to their risks.

Machine learning can take care of at least some of this shortfall since the model trained on the historic information can factor in several things at once and use these rules to apply to present-day students. Machine learning has been used before for the behavioral monitoring and prediction purposes [1]. Yet, for a functional system to help students, one needs more than a classifier. There has to be an integration of the behavioral and academic data, a split between prediction of the performance and of the likelihood of dropout, a numerical representation of the outcome to be ranked by the professors, as well as reasons for flagging a student.

Dropout examples are, more often than not, very small in number. In such cases, a machine learning model can achieve high overall accuracy levels just by successfully predicting the major class and miss out on a huge number of students who are in fact at risk of dropping out. This can lead to a false impression of the model's effectiveness since the students that should be given attention are incorrectly labeled as low risk. For this reason, accuracy alone is not sufficient for measuring the dropout prediction efficiency. It is vital to look at such metrics as precision, recall, and F1-score as well in order to get a better understanding of the operations of the model.

Precision, recall and F1-score therefore need to be examined together with accuracy.

In this paper, EDU AI is presented as the decision-support system developed to meet all those criteria. It will involve gathering and verifying data of students, performing data pre-processing and feature engineering, using four machine learning algorithms in their specific positions, transforming the results into the individual risk score with three levels: Low, Medium and High, as well as displaying the trends, risk distribution and the list of students under the risk on an analytical dashboard. Predictions are supposed to be used by authorized faculty and administrators as the source of decision-making, but not as a substitute for academic judgement.

There are four research questions that lead the research work. The first research question, namely RQ1, looks into whether it is possible to use academic information, attendance record and behavioural information to predict the performance and the probability of dropout using machine learning algorithms. The second research question, RQ2, focuses on the performances of Logistic Regression, Random Forest, Naïve Bayes and K-Means in predicting the student performance in their respective roles based on the available student information. The third research question, RQ3, seeks to establish whether it is possible to present the results in the form of a risk score, classification of three levels and contributors.

II. RELATED WORK

A. Educational Data Mining and Performance Prediction

Educational Data Mining (EDM) employs data mining and statistical methods to analyze the data generated in educational environments, like attendance data, test data, and learning management system interactions, to understand why certain students perform well and others do not [2], [3]. The most common applications include classification of students, clustering of similar students, and prediction of performance. Performance prediction research relies on attributes like past grades, attendance, test scores, and learning behavior to predict marks or grades through classification or regression techniques. Cortez and Silva [4], for instance, employed binary and five-level classification and regression of secondary school performance based on school reports and questionnaires, finding that high levels of accuracy could be obtained when the earlier period grades were known, despite other features also being relevant. One common issue with these research efforts is their reliance on only one course or institution. *Dropout Prediction and Class Imbalance*

Typically, the dropout prediction problem is approached using binary classification, where it is determined whether or not a student will drop out of the program. Academic, demographic and behavioral factors are often considered simultaneously by researchers. As a result of the underrepresentation of dropout cases in data sets, the class imbalance problem can cause a classifier to seem

which is why other evaluation metrics apart from accuracy should be considered. For example, Dekker et al. [5] used decision tree techniques to predict engineering freshmen's dropout and focused on cost-sensitive learning and misclassification analysis. Thus, EDU AI treats dropout prediction separately from performance prediction and provides several evaluation metrics alongside with accuracy.

B. Algorithms, Early Warning Systems and Risk Scoring

Methods like Logistic Regression, Random Forest, Naïve Bayes and K-Means Clustering are popular when working with education data sets. Logistic Regression and Naïve Bayes are appreciated for simplicity, speed when dealing with small sets of data and interpretability. Random Forest, which uses a number of decision trees, better reflects the non-linear relations and interactions between features while K-Means is an unsupervised technique for clustering similar students. As there is no universal algorithm applicable everywhere, researchers usually test a couple of different algorithms on the same dataset; for example, one study analyzing dropouts included Decision Trees, Logistic Regression, Naïve Bayes and Random Forest [6]. The early warning system highlights problematic students in advance on the basis of small set of factors like attendance and performance in assessments. While the rule-based system is simple and understandable, it does not reflect the joint impact of all factors, while a predictive system reflects it but must be presented in a simple form understandable for teachers. Using scoring or categorization system allows highlighting most problematic students on the basis of a score description.

C. Dashboards, Explainability and Responsible Use

Research in learning analytics emphasizes the use of dashboards to display information like student attendance, assessment results and risk levels to educators and administrators alike, and makes the argument that dashboards that provide good information lead to a decision or action. Predictive analytics in education has also raised issues like privacy of information, equity among different groups of students, labelling of students, and need for transparency in models, which indicates how predictions should be used for making decisions, rather than being used automatically. The related information on explainable AI in predictive analytics [7] and principles of AI-based analytics beyond automation [8] is consistent with that.

Most of the existing researches primarily concentrate on one aspect, such as focusing either on one type of prediction task, an algorithm comparison, or a dashboard. What is still missing is one unified faculty-centered system that connects validated student data, data preprocessing, dropout prediction as well as performance and individual risk scoring.

EDU AI is designed to meet this need.

accurate, yet miss a large number of dropout students,

III. PROPOSED EDU AI FRAMEWORK

A. System Architecture and Data Flow

EDU AI's process can be viewed as following a pipeline from data collected from students through to the decisions that faculty make (Fig. 1). All student data is established by an authenticated user and stored in the student database. The data is first pre-processed, resulting in data that has been cleaned, processed for missing values, categorical variables transformed and numeric values integrated. The features are then selected in the feature engineering stage of the process and reliable features are chosen. A machine learning engine is then employed to perform two tasks from this set of features. The dropout stage follows soon after, in which dropout risk is evaluated, using the correct features as well as predicted performance. Risk assessments, as well as classifications of risk, are completed and this information is presented at the analytics dashboard. In the whole process the integrity of the respective data is maintained and all information that is outputted has been comprehensively put together and accepted. Raw and processed student data have been kept distinctly separate, until the time that the final scores have been calculated.

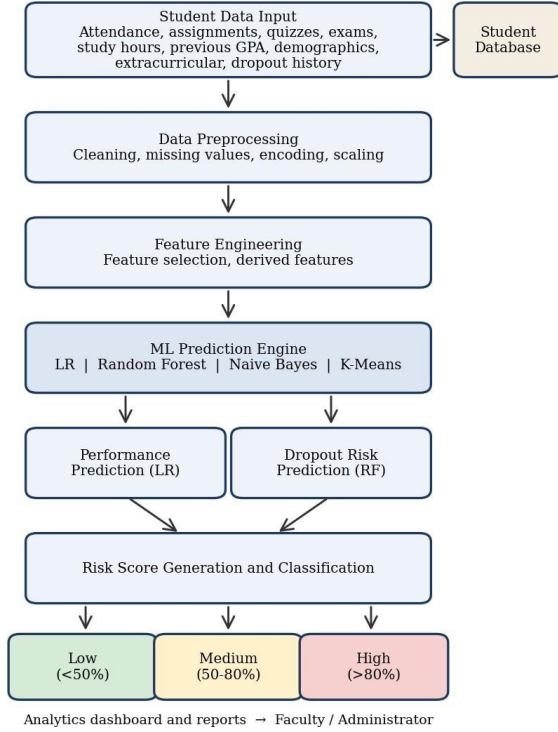


Fig. 1. EDU AI system architecture.

B. Student Data and Validation

Each student is identified by a unique identifier to which all academic, behavioural and historical records are linked. The data categories are listed in Table I. Before storage the

data is validated: attendance and scores must lie in valid ranges, study hours must be non-negative, identifiers must be unique and mandatory fields must be present. Records that fail validation are returned to the user with a message for correction. The database is organised around the student entity, with entities for academic records, behavioural data, predictions, risk scores and analytics summaries, so that results can be retained over time for trend analysis.

TABLE I. STUDENT DATA CATEGORIES USED BY EDU AI

Category	Representative features	Purpose
Student information	Student ID, name, age, gender, department	Identifies the student and links records
Attendance	Attendance records or percentage	Regularity and engagement
Assessment data	Assignment scores, quiz scores, examination marks	Academic achievement
Previous record	Previous GPA	Past academic standing
Study behaviour	Study hours	Academic effort
Demographics	Demographic attributes	Background context
Extracurricular	Participation in activities	Wider engagement
Historical records	Historical dropout records	Outcomes for model training

C. Preprocessing and Feature Engineering

Raw student data may contain missing values, inconsistent formats, categorical attributes and values on different scales. The preprocessing stage removes duplicates, corrects inconsistent formats and handles invalid or outlying values. Missing attendance, scores or other fields are treated by imputation or by flagging the record, depending on the field, so that incomplete records need not be discarded. Categorical attributes such as gender, department and extracurricular participation are encoded numerically, and numerical features are scaled where the algorithm is sensitive to magnitude, as with Logistic Regression. Feature engineering then derives measures that describe behaviour more directly, for example an overall assessment average, the trend of quiz or assignment scores across the term, the change in GPA and the relationship between study hours and results. The final feature set is split into training and testing portions for model development. The features finally retained are decided during model development.

D. Machine Learning Models

Four algorithms are used, each with a specific role (Table II). Logistic Regression computes the performance score by passing a weighted linear combination of the student features through the sigmoid function and scaling the result to a percentage:

$$z = \sum \beta_i x_i + \beta_0, \sigma(z) = 1 / (1 + e^{-z}), \text{Score\%} = \sigma(z) \times 100 \quad (1)$$

where β_0 is the intercept and β_i are coefficients learned from the data. Each coefficient shows how strongly a feature affects the score, which makes the model useful for

explaining results to faculty. Random Forest classifies the dropout risk class by majority vote of its trees:

$$RF(x) = \operatorname{argmax}_c \sum_k I(T_k(x) = c) \quad (2)$$

where T_k is the k -th tree and I is an indicator function that equals 1 when the tree votes for class c . The implemented forest contains 10 decision trees trained on random feature subsets (bagging). Naïve Bayes assigns a student to the class with the highest posterior probability, assuming Gaussian likelihoods for continuous features and class priors from the dataset distribution:

$$P(c | X) \propto P(c) \cdot \prod_i P(x_i | c) \quad (3)$$

K-Means partitions the students into $K = 4$ clusters by minimising the within-cluster sum of squared Euclidean distances, assigning each student to the nearest centroid:

$$\operatorname{cluster}(x) = \operatorname{argmin}_k \|x - \mu_k\|^2 \quad (4)$$

Pre-converged centroids are used for reproducibility. The training phase learns the models from historical student records and compares their performance, and the prediction phase applies the selected trained models to current student data.

TABLE II. MACHINE LEARNING MODELS AND THEIR ROLES

Model	Description	Role in EDU AI
Logistic Regression	Weighted linear combination of features passed through a sigmoid; coefficients are directly interpretable	Performance prediction
Random Forest	Ensemble of 10 decision trees (bagging); majority vote decides the class	Dropout risk classification
Naïve Bayes	Bayes' theorem with Gaussian likelihoods; efficient on small datasets	Behavioural risk assessment
K-Means	Unsupervised, $K = 4$ clusters, pre-converged centroids	Student segmentation

E. Risk Score, Risk Levels and Contributing Factors

The risk score is an individual value, expressed as a percentage from 0 to 100, that summarises the performance prediction and the dropout risk estimate.

Conceptually it is a function of the model outputs and the main student indicators:

$$\text{Risk Score} = f(\text{performance, dropout estimate, indicators}) \quad (5)$$

The components are academic performance, attendance, study behaviour, dropout probability and background factors. The exact function f and the weight of each component are not detailed in this paper. The score is mapped to three levels (Table III) so that faculty can see at a glance which students need attention. For every flagged student the system also identifies the factors that contributed most, such as low attendance, declining quiz scores, reduced study hours or a low previous GPA, using model outputs

such as feature importance or coefficients. The result is stored with the student record so that changes in risk can be observed over time.

TABLE III. RISK LEVEL CLASSIFICATION

Risk level	Risk score	Interpretation and suggested faculty attention
Low (Stable)	< 50%	No major warning signs; routine monitoring
Medium (Moderate)	50–80%	Some warning signs that may worsen; closer monitoring and early academic support
High (Critical)	> 80%	Several strong warning signs; priority attention and timely intervention

F. Analytics Dashboard and Reporting

The dashboard reads the stored data, predictions and risk scores and presents student performance trends, attendance analytics, risk distribution, an at-risk list ordered by risk score, and prediction results with contributing factors. A dropout risk analysis view shows counts of students in the Stable, Moderate and Critical categories with filters by risk level and department, a card per student showing attendance, GPA (10-point scale), performance and risk level, and actions to intervene, export the list and run batch analysis. An ML models explorer compares the four algorithms by accuracy, precision, recall and F1-score, shows the confusion matrix and feature importances of a selected model, and displays the K-Means clusters. A reporting function summarises performance, risk levels, at-risk students and trends for academic review. Access to dashboards and reports is restricted to authorised users because they contain sensitive information about individual students.

IV. EXPERIMENTAL METHODOLOGY

A. Dataset and Evaluation Scope

The evaluation uses a dataset of 248 students, and the results are those reported by the ML models explorer of the implemented system. It contains academic and behavioural attributes for each student and was used for student performance and dropout-risk analysis. The data were divided into training and testing portions for model development and evaluation; the split ratio and the original source of the data are not reported here. The dataset is strongly imbalanced across the risk classes: 207 students are Low risk, 29 Medium risk and 12 High risk. The study is therefore treated as a feasibility evaluation of the integrated system, and the reported figures should not be read as estimates of performance at other institutions.

B. Models and Configuration

Logistic Regression produces the performance score, a Random Forest of 10 trees produces the dropout risk class (Low, Medium or High), Naïve Bayes with Gaussian likelihoods supports behavioural risk assessment, and K-Means with $K = 4$ provides segmentation. The system is

implemented in Python, using Pandas and NumPy for data handling, Scikit-learn for model development, and Matplotlib and Seaborn for visualisation.

Hyperparameters beyond those stated are not reported here.

C. Evaluation Metrics

The supervised models are evaluated with accuracy, precision, recall and F1-score. Accuracy is the proportion of correctly classified samples among all samples evaluated, and precision, recall and F1-score are calculated from the predicted and actual class labels. For a class, with true positives TP, false positives FP and false negatives FN:

$$\text{Precision} = TP / (TP + FP), \quad (6)$$

$$\text{Recall} = TP / (TP + FN) \quad (7)$$

$$F1 = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall}) \quad (7)$$

Recall is emphasised for the at-risk classes because failing to detect an at-risk student is usually more costly than monitoring an additional student. The Random Forest confusion matrix over the three risk classes is also reported. As a supplementary analysis, per-class precision, recall and F1-score and a combined at-risk (Medium or High) detection rate were computed directly from that confusion matrix. These derived values are labelled as such and are not outputs of the system. K-Means is unsupervised, so its evaluation differs from that of the supervised models: its four reported values are taken as output by the system, and the procedure used to compute them.

V. RESULTS

A. Model Performance

Table IV and Fig. 2 compare the four models. Random Forest achieved the highest value on all four measures (accuracy 94.8%, precision 92.1%, recall 94.5%, F1-score 93.3%). Logistic Regression was second (accuracy 91.2%, F1-score 90.0%), followed by K-Means (88.6%) and Naïve Bayes (87.4%). Because the models serve different purposes, the comparison shows how well each performs in its assigned role rather than a ranking of interchangeable classifiers.

TABLE IV. PERFORMANCE OF THE MACHINE LEARNING MODELS

Model	Role	Acc.	Prec.	Rec.	F1
Logistic Regression	Performance	91.2%	89.7%	90.4%	90.0%
Random Forest	Dropout	94.8%	92.1%	94.5%	93.3%
Naïve Bayes	Behavioural risk	87.4%	86.1%	88.2%	87.1%
K-Means	Segmentation	88.6%	87.3%	89.1%	88.2%

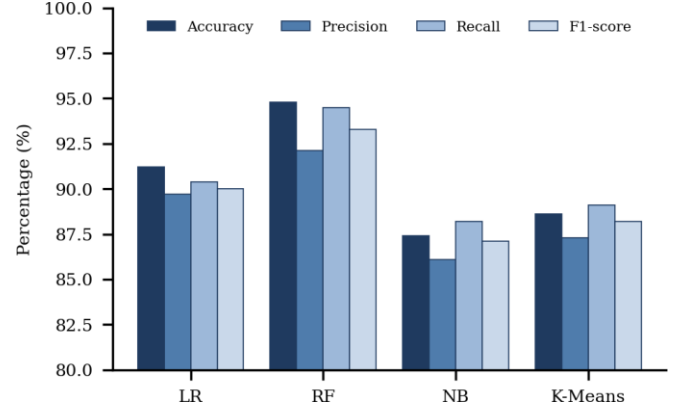


Fig. 2. Model performance comparison (LR: Logistic Regression, RF: Random Forest, NB: Naïve Bayes).

B. Dropout Risk Classification

Table V gives the Random Forest confusion matrix. Of the 207 Low-risk students, 201 were predicted correctly; of the 29 Medium-risk students, 24 were correct; and of the 12 High-risk students, 10 were correct. The most serious error, a High-risk student predicted as Low risk, occurred once.

TABLE V. CONFUSION MATRIX OF THE RANDOM FOREST CLASSIFIER

Actual \ Predicted	Low	Medium	High
Low risk	201	4	2
Medium risk	3	24	2
High risk	1	1	10

Table VI lists per-class values computed from Table V. Recall is 97.1% for Low risk, 82.8% for Medium risk and 83.3% for High risk, and precision for the High class is 71.4%, so the small at-risk classes are identified less reliably than the majority class. Treating Medium and High together as “at risk”, 37 of the 41 at-risk students were flagged (recall 90.2%) and 37 of the 43 flagged students were truly at risk (precision 86.0%). The matrix gives an overall accuracy of $235/248 = 94.8\%$, which is the Random Forest accuracy used in this paper. The precision, recall and F1-score of 92.1%, 94.5% and 93.3% in Table IV are as output by the system’s explorer and cannot be reproduced from Table V.

TABLE VI. PER-CLASS METRICS DERIVED FROM TABLE V

Class	Precision	Recall	F1-score
Low risk	98.0%	97.1%	97.6%
Medium risk	82.8%	82.8%	82.8%
High risk	71.4%	83.3%	76.9%
Macro average	84.1%	87.7%	85.8%

C. Risk Level Distribution

Applying the thresholds of Table III to the 248 students gives the distribution in Table VII and Fig. 3: 207 students (83.5%) are Low risk, 29 (11.7%) Medium risk and 12 (4.8%) High risk. In total 41 students are flagged for attention, 12 of them in the critical category. This is the list that the dashboard presents to faculty, ordered by risk score.

TABLE VII. STUDENT RISK LEVEL DISTRIBUTION

Risk level	Students	Percentage
Low (Stable, < 50%)	207	83.5%
Medium (Moderate, 50–80%)	29	11.7%
High (Critical, > 80%)	12	4.8%

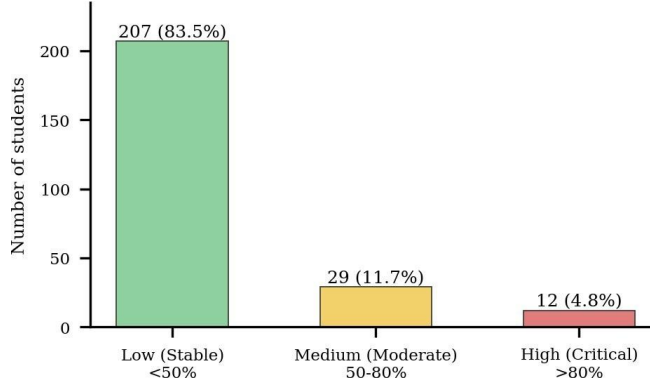


Fig. 3. Distribution of students across risk levels.

D. Implemented Functionality

The dashboard views described in Section III-F, including the dropout risk analysis, the at-risk list with filters and the ML models explorer with confusion matrix, feature importances and K-Means clusters, are part of the implemented system from which the results were taken. The effect of faculty interventions on student outcomes has not been measured, so the intervention support offered by the system is a capability rather than an evaluated result.

VI. DISCUSSION

A. Roles of the Four Models

By assigning each algorithm a separate role, it makes it easier to interpret the produced results. Logistic Regression is a perfect machine in terms of evaluating its accuracy since the coefficients corresponding to each variable demonstrate the level of effect exerted on the outcome variable. Random Forest proved to be the most accurate one among all approaches and also yielded the highest percentage of true positive outcomes, which makes it the most adequate for performing dropout prediction, since the cost of misidentifying an at-risk student is higher than that of unnecessary observation of a student. Naïve Bayes and K-Means, on their part, achieved lower scores and have been utilized as supportive methods. Nevertheless, both these methods require particular caution.

B. Imbalance and the Cost of Missed Students

The class distribution (207, 29 and 12 students) shows why accuracy alone is insufficient. A classifier that labelled every student Low risk would still be correct for 83.5% of students while detecting none of the students who need help. With only 12 High-risk students, a single misclassification changes the High-class results noticeably, which is visible in the 71.4% precision and 83.3% recall derived from Table V. The one High-risk student predicted as Low risk is the main reason recall for the at-risk classes should be

monitored alongside overall accuracy, and a larger dataset is needed to estimate these rates with useful confidence.

C. Interpretability and Decision Support

The risk score, the three levels and the listed contributing factors are designed so that a faculty member can see who needs attention and why. A High-risk label indicates that attention is advisable and does not imply that a student will certainly fail or leave. This framing is consistent with concerns in the literature about labelling students and about transparency [7], and it keeps the final judgement with the educator. Because the weighting of the score components is not detailed here, the paper cannot state how strongly each component influences a level.

D. Design Recommendations

Several design points follow from this work. Performance prediction and dropout prediction should remain separate tasks. Evaluation should report precision, recall and F1-score per class together with accuracy. Risk thresholds and score components should be documented and explained to users. Raw data should be kept separate from derived predictions so that results remain consistent and traceable. Finally, access to predictions should be limited to authorised users, since they describe individual students.

VII. LIMITATIONS AND FUTURE WORK

A. Limitations

The evaluation rests on 248 students from a single dataset, and only 12 of them are High risk, so the results may not generalise across programmes, institutions or academic years. The original source of the data, the train–test split ratio and the K-Means evaluation procedure are not documented, and the Random Forest precision, recall and F1-score reported by the system cannot be reproduced from the confusion matrix. The confusion matrix covers all 248 students, so it is not clear that it reflects held-out data. The risk score formula and weights are not specified. No fairness analysis across student groups, no temporal validation across semesters and no study of intervention effectiveness were performed. The results are those reported by the system’s own explorer and have not been independently replicated. They should therefore be read as evidence of feasibility, not as a field-validated estimate of performance.

B. Future Work

The following extensions are proposed and have not been implemented: training on data from more students, programmes and academic years; real-time monitoring that updates predictions as new attendance and assessment data arrive; evaluation of deep learning where sufficient data exist; stronger explainable-AI support; integration with college ERP and LMS systems; a mobile application; richer visualisation; automated alerts to faculty when risk rises; periodic retraining with fairness checks; and long-term

tracking of student progress and intervention outcomes. All future development should preserve the privacy of student data, restrict access to authorised users and keep predictions explainable and used as decision support.

VIII. CONCLUSION

This paper presented EDU AI, an AI-based Student Performance & Dropout Prediction System that supports early identification of students at risk of poor academic performance or dropout. The system validates and stores student academic and behavioural data, prepares features, applies Logistic Regression, Random Forest, Naïve Bayes and K-Means in distinct roles, generates an individual risk score with Low, Medium and High levels and contributing factors, and presents the results through an analytics dashboard and reports. On 248 students, Random Forest gave the best reported results (accuracy 94.8%, recall 94.5%, F1-score 93.3%), the system flagged 41 students including 12 in the critical category, and one High-risk student was predicted as Low risk. These results show that the integrated approach is feasible, but they come from a small, imbalanced dataset with undocumented validation and no measurement of intervention outcomes. Validation on larger, multi-term data and documentation of the risk score formula are the next steps before operational use.

The system remains a decision-support tool in which predictions guide faculty and the final judgement stays with the educator.

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